

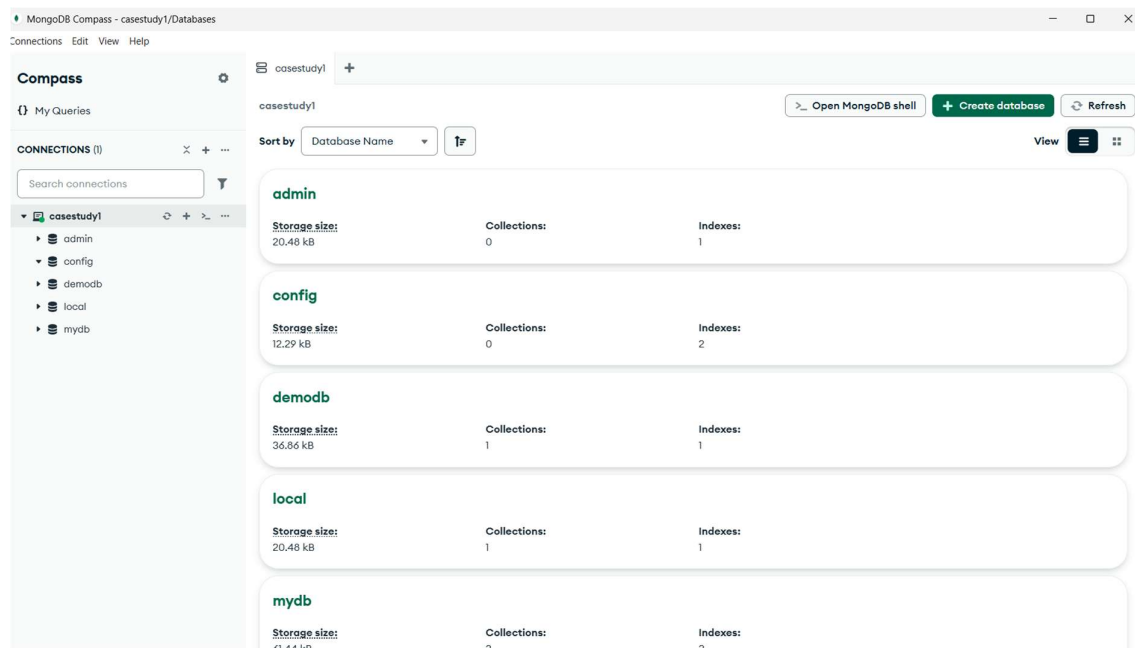
MongoDB Product Collection Case Study - Documentation

1. Opening MongoDB Compass

Step:

I launched **MongoDB Compass** from my system to begin managing the product collection.

Screenshot:



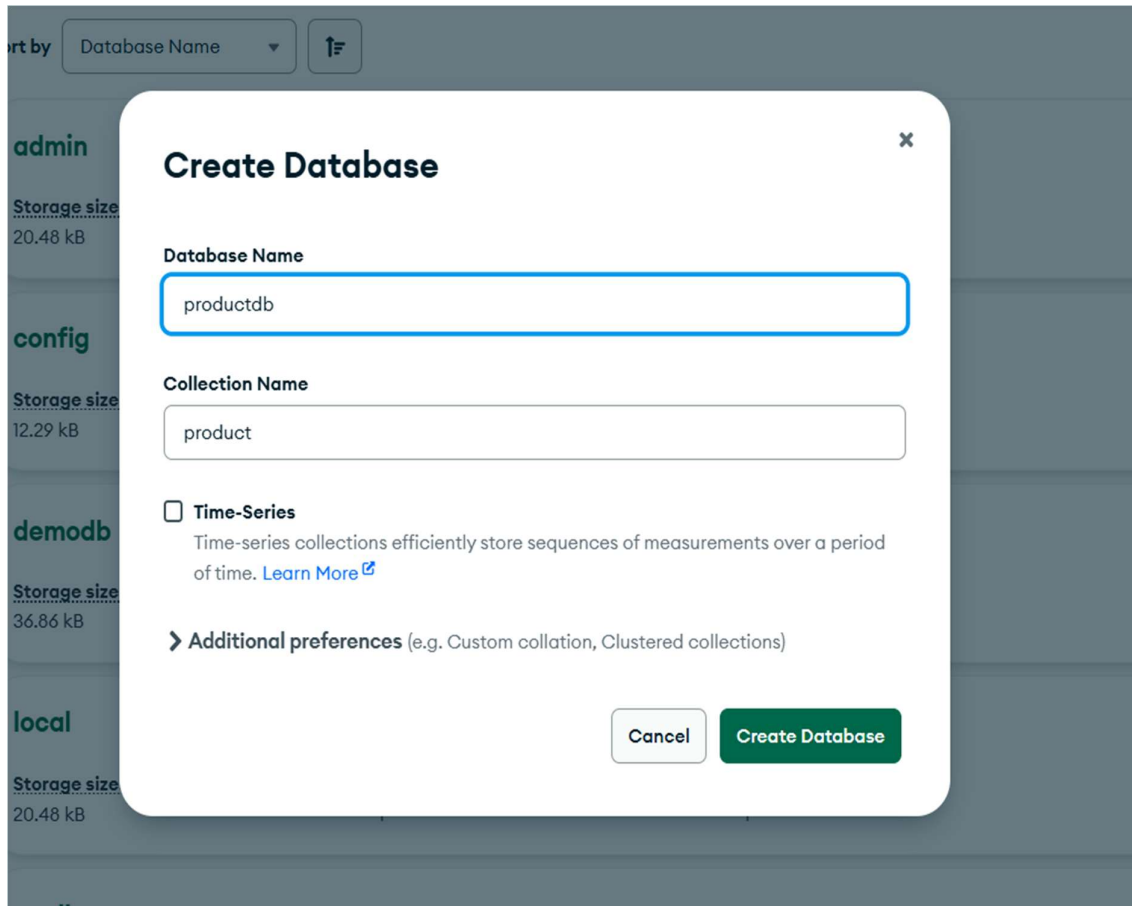
2. Creating Database and Collection

Step:

I created a new database and collection by clicking on "**Create Database**".

- **Database Name:** productdb
- **Collection Name:** product

 Screenshot:



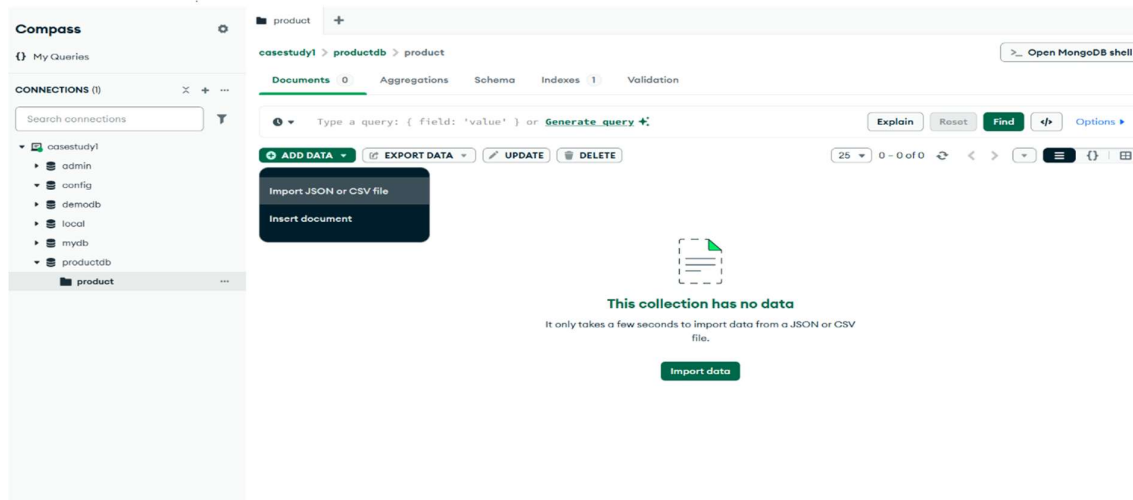
 3. Inserting JSON Data into the Collection

Step:

I selected the product collection and clicked **“Insert Document”** to add sample product data.

Instead of manually entering documents, I used the **“Import”** option to add a JSON file containing multiple product documents.

Screenshot:

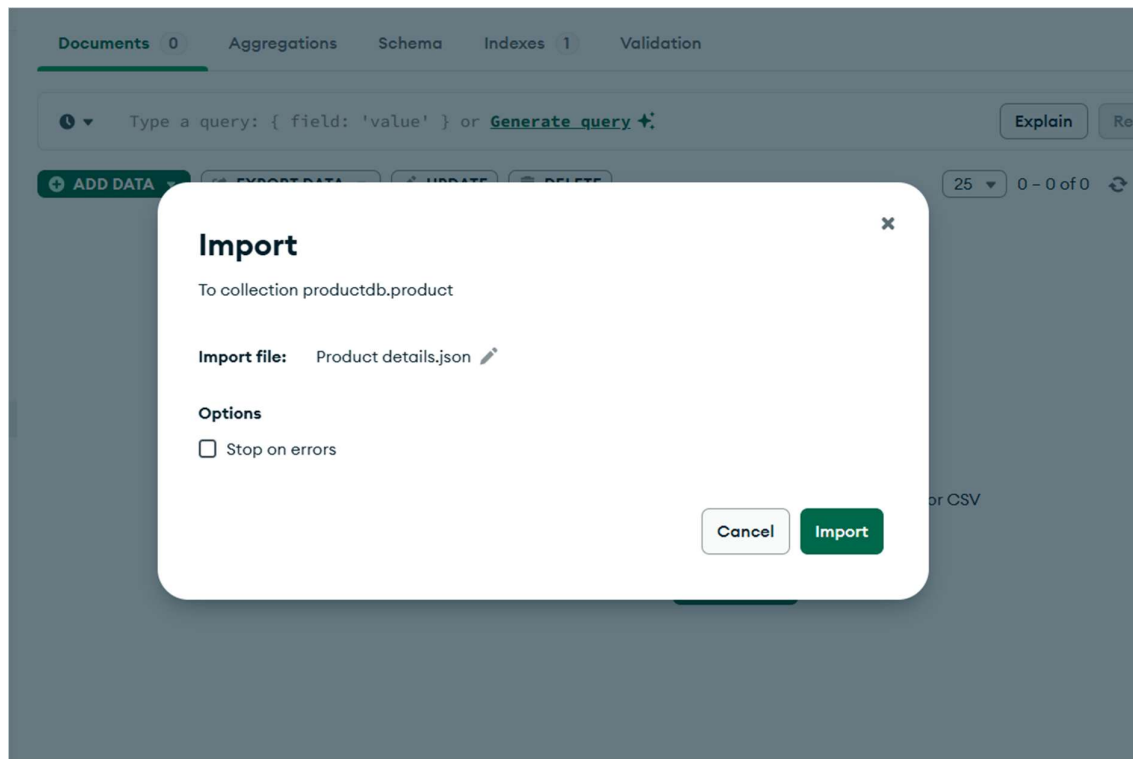


4. Using Import Option to Add Bulk Documents

Step:

I selected the “**Import**” button, chose the JSON file, and uploaded all 20 product documents into the collection.

Screenshot:

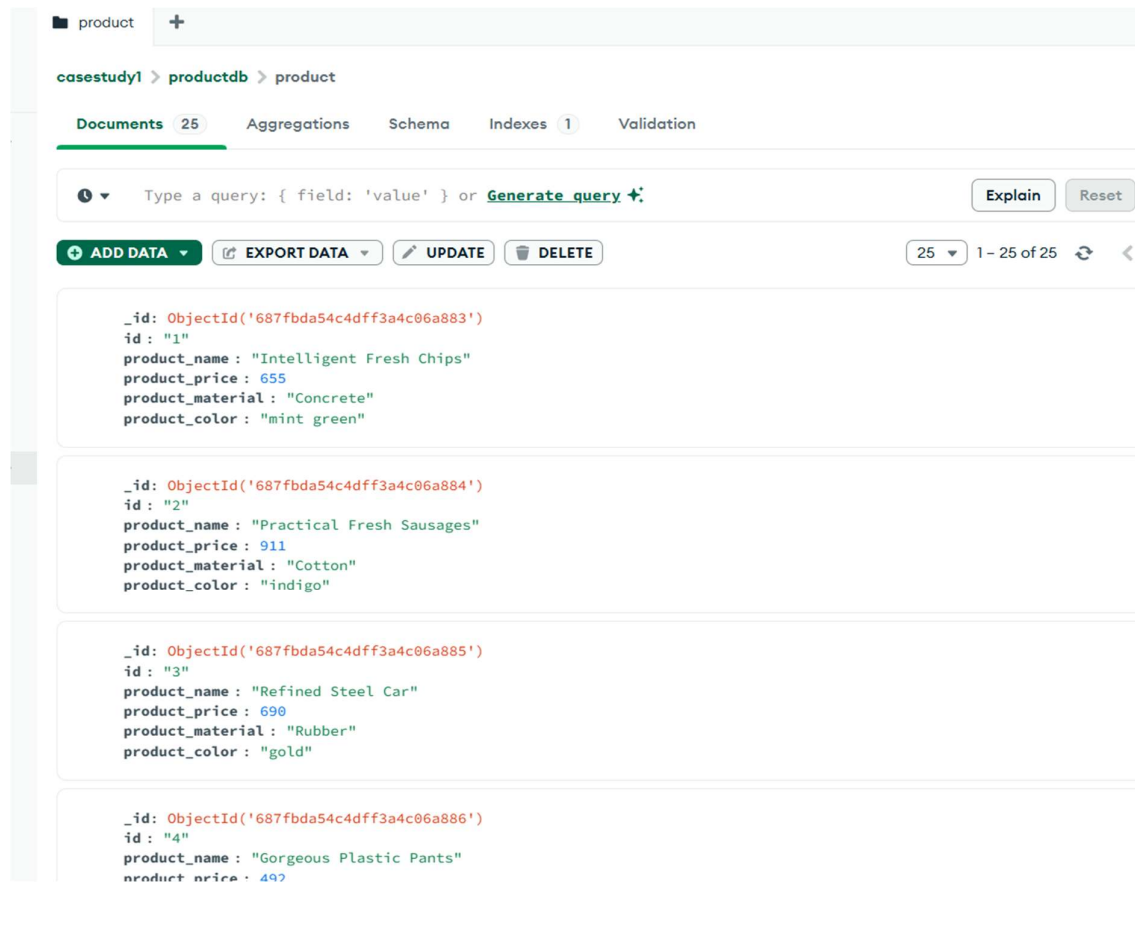


📌 5. Viewing Inserted Documents in MongoDB Compass UI

Step:

After import, I selected the product collection and verified that the product documents were added successfully.

🖼️ Screenshot:



📌 6. Opening Mongo Shell in Compass

Step:

I switched to the **Mongo Shell tab** within Compass to execute queries manually. To verify my current collection, I used:

use productdb

db.product.find()

Screenshot:

```
>_MONGOOSH
> use productdb
< switched to db productdb
> db.product.find()
< {
  _id: ObjectId('687fbda54c4dff3a4c06a883'),
  id: '1',
  product_name: 'Intelligent Fresh Chips',
  product_price: 655,
  product_material: 'Concrete',
  product_color: 'mint green'
}
{
  _id: ObjectId('687fbda54c4dff3a4c06a884'),
  id: '2',
  product_name: 'Practical Fresh Sausages',
  product_price: 911,
  product_material: 'Cotton',
  product_color: 'indigo'
}
{
  _id: ObjectId('687fbda54c4dff3a4c06a885'),
  id: '3',
  product_name: 'Refined Steel Car',
  product_price: 690,
  product_material: 'Rubber',
  product_color: 'gold'
}
{
```

7. Executing Queries with Explanation

Query 1: Find all the information about each product

db.product.find()

Explanation: Retrieves all documents in the product collection.

Screenshot of output

```
product  mongosh: oasestudy1  +
>_MONGOSH
> db.product.find()
< {
  _id: ObjectId('687fbda54c4dff3a4c06a883'),
  id: '1',
  product_name: 'Intelligent Fresh Chips',
  product_price: 655,
  product_material: 'Concrete',
  product_color: 'mint green'
}
{
  _id: ObjectId('687fbda54c4dff3a4c06a884'),
  id: '2',
  product_name: 'Practical Fresh Sausages',
  product_price: 911,
  product_material: 'Cotton',
  product_color: 'indigo'
}
{
  _id: ObjectId('687fbda54c4dff3a4c06a885'),
  id: '3',
  product_name: 'Refined Steel Car',
  product_price: 690,
  product_material: 'Rubber',
  product_color: 'gold'
}
{
  _id: ObjectId('687fbda54c4dff3a4c06a886'),
  id: '4',
```

◆ Query 2: Find products with price between 400 and 800

```
db.product.find({ product_price: { $gte: 400, $lte: 800 } })
```

Explanation: Filters products where price is between ₹400 and ₹800.

Screenshot of output

```
> db.product.find({ product_price: { $gte: 400, $lte: 800 } })
< {
  _id: ObjectId('687fbda54c4dff3a4c06a883'),
  id: '1',
  product_name: 'Intelligent Fresh Chips',
  product_price: 655,
  product_material: 'Concrete',
  product_color: 'mint green'
}
{
  _id: ObjectId('687fbda54c4dff3a4c06a885'),
  id: '3',
  product_name: 'Refined Steel Car',
  product_price: 690,
  product_material: 'Rubber',
  product_color: 'gold'
}
{
  _id: ObjectId('687fbda54c4dff3a4c06a886'),
  id: '4',
  product_name: 'Gorgeous Plastic Pants',
  product_price: 492,
  product_material: 'Soft',
  product_color: 'plum'
}
```

```

{
  _id: ObjectId('687fbda54c4dff3a4c06a886'),
  id: '4',
  product_name: 'Gorgeous Plastic Pants',
  product_price: 492,
  product_material: 'Soft',
  product_color: 'plum'
}
{
  _id: ObjectId('687fbda54c4dff3a4c06a888'),
  id: '6',
  product_name: 'Awesome Wooden Towels',
  product_price: 474,
  product_material: 'Plastic',
  product_color: 'orange'
}
{
  _id: ObjectId('687fbda54c4dff3a4c06a889'),
  id: '7',
  product_name: 'Practical Soft Shoes',
  product_price: 500,
  product_material: 'Rubber',
  product_color: 'pink'
}

```

◆ Query 3: List four products with price greater than 500

```
db.product.find({ product_price: { $gt: 500 } }).limit(4)
```

Explanation: Shows first four products priced above ₹500.

 *Screenshot of output*

```

>_MONGOSH
>
Type "it" for more
> db.product.find({ product_price: { $gt: 500 } }).limit(4)
< {
  _id: ObjectId('687fbda54c4dff3a4c06a883'),
  id: '1',
  product_name: 'Intelligent Fresh Chips',
  product_price: 655,
  product_material: 'Concrete',
  product_color: 'mint green'
}
{
  _id: ObjectId('687fbda54c4dff3a4c06a884'),
  id: '2',
  product_name: 'Practical Fresh Sausages',
  product_price: 911,
  product_material: 'Cotton',
  product_color: 'indigo'
}
{
  _id: ObjectId('687fbda54c4dff3a4c06a885'),
  id: '3',
  product_name: 'Refined Steel Car',
  product_price: 690,
  product_material: 'Rubber',
  product_color: 'gold'
}

```

◆ **Query 4: Find product with row id = "10"**

```
db.product.find({ id: '10' })
```

Explanation: Finds the product with string id "10".

 *Screenshot of output*

```
> db.product.find({ id: '10' })
< {
  _id: ObjectId('687fbda54c4dff3a4c06a88c'),
  id: '10',
  product_name: 'Generic Wooden Pizza',
  product_price: 84,
  product_material: 'Frozen',
  product_color: 'indigo'
}
productdb>
```

◆ **Query 5: Find products with material = "Soft"**

```
db.product.find({ product_material: 'Soft' })
```

Explanation: Filters products where material is 'Soft'.

 *Screenshot of output*

```
> db.product.find({ product_material: 'Soft' })
< [
  {
    _id: ObjectId('687fbda54c4dff3a4c06a886'),
    id: '4',
    product_name: 'Gorgeous Plastic Pants',
    product_price: 492,
    product_material: 'Soft',
    product_color: 'plum'
  },
  {
    _id: ObjectId('687fbda54c4dff3a4c06a88b'),
    id: '9',
    product_name: 'Awesome Wooden Ball',
    product_price: 28,
    product_material: 'Soft',
    product_color: 'azure'
  },
  {
    _id: ObjectId('687fbda54c4dff3a4c06a88d'),
    id: '11',
    product_name: 'Unbranded Wooden Cheese',
    product_price: 26,
    product_material: 'Soft',
    product_color: 'black'
  }
]
```

✅ Conclusion

In this MongoDB task, I:

- Successfully created a database and collection in MongoDB Compass.
- Inserted JSON documents using the Import feature.
- Accessed the Mongo Shell and wrote multiple queries.
- Filtered documents based on price, material, and other conditions.

This activity helped me understand practical MongoDB CRUD operations and enhanced my confidence in NoSQL data handling using Compass and Shell.
